

Structural and Dynamic Impacts of Single-atom Disruptions to Guide RNA Interactions within the Recognition Lobe of *Geobacillus stearothermophilus* Cas9

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This study offers **valuable** insights into the conformational dynamics of the nucleic acid recognition lobe of GeoCas9, a thermophilic Cas9 from *Geobacillus stearothermophilus*. The authors investigate the influence of local dynamics and allosteric regulation on guide RNA binding affinity and DNA cleavage specificity through molecular dynamics simulations, advanced NMR techniques, RNA binding studies, and mutagenesis. While the mutations studied do not lead to significant changes in GeoCas9 cleavage activity, the study provides **convincing** evidence for the role of allosteric mechanisms and interdomain communication in Cas9 enzymes, and will be of great interest to biochemists and biophysicists exploring these complex systems.

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Abstract

The intuitive manipulation of specific amino acids to alter the activity or specificity of CRISPR-Cas9 has been a topic of great interest. As a large multi-domain RNA-guided endonuclease, the intricate molecular crosstalk within the Cas9 protein hinges on its conformational dynamics, but a comprehensive understanding of the extent and timescale of the motions that drive its allosteric function and association with nucleic acids remains elusive. Here, we investigated the structure and multi-timescale molecular motions of the recognition (Rec) lobe of GeoCas9, a thermophilic Cas9 from *Geobacillus stearothermophilus*. Our results provide new atomic details about the GeoRec subdomains (GeoRec1, GeoRec2) and the full-length domain in solution. Two rationally designed mutants, K267E and R332A, enhanced and redistributed micro-millisecond flexibility throughout GeoRec, and NMR studies of the interaction between GeoRec and its guide RNA showed that mutations reduced

this affinity and the stability of the ribonucleoprotein complex. Despite measured biophysical differences due to the mutations, DNA cleavage assays reveal no functional differences in on-target activity, and similar specificity. These data suggest that guide RNA interactions can be tuned at the biophysical level in the absence of major functional losses but also raise questions about the underlying mechanism of *GeoCas9*, since analogous single-point mutations have significantly impacted on- and off-target DNA editing in mesophilic *S. pyogenes* Cas9. A K267E/R332A double mutant did also did not enhance *GeoCas9* specificity, highlighting the robust tolerance of mutations to the Rec lobe of *GeoCas9* and species-dependent complexity of Rec across Cas9 paralogs. Ultimately, this work provides an avenue by which to modulate the structure, motion, and guide RNA interactions at the level of the Rec lobe of *GeoCas9*, setting the stage for future studies of *GeoCas9* variants and their effect on its allosteric mechanism.

Introduction

The vast majority of Cas systems explored as genome editors originate from mesophilic hosts. The emergence of the thermophilic *GeoCas9*, with DNA cleavage function up to 85°C, can expand CRISPR technology to higher temperature regimes and stabilities,^{1,2} but its regulatory mechanism relative to canonical Cas9s must be established. The *SpyCas9*, which originates from the mesophilic *Streptococcus pyogenes*, as well as *GeoCas9*, are both effectors of Type-II CRISPR systems. Interestingly, the Type II-A *SpyCas9* has been by far the most used Cas enzyme, including in ongoing clinical trials.^{3,4} But Cas9 homologs of the Type II-C class, such as *Neisseria meningitis* (*NmeCas9*) and *Campylobacter jejuni* (*CjeCas9*), to which *GeoCas9* belongs, have been validated for mammalian genome editing,^{2,5,6} reinforcing the need to better understand this CRISPR class.

The similar domain arrangements of *GeoCas9* and *SpyCas9* led us to initially speculate that these could share atomic level mechanistic similarities.⁷ *GeoCas9* utilizes a guide RNA (gRNA) to localize and unwind a double-stranded DNA (dsDNA) target after recognition of its 5'NNNNCRAA-3' protospacer adjacent motif (PAM).^{2,8} Upon recognition of the PAM sequence by the PAM-Interacting domain (PI), Cas9-bound guide (gRNA) forms an RNA:DNA hybrid with the target DNA strand. Initially thought to be part of the PI domain², the wedge (WED) domain recognizes the repeat:anti-repeat region of the gRNA and the dsDNA upstream of the target region.⁹ The Rec lobe of Cas9 is responsible for orienting the RNA:DNA hybrid, as well as the adjacent nuclease domains, into their active conformations.^{10–13} Coordinated cleavage of the target and non-target DNA strand then occurs via the HNH and RuvC nucleases, respectively. The *GeoCas9* nuclease active sites within HNH and RuvC are spatially distinct from the PAM recognition site in the PI domain, necessitating structural and dynamic changes that allosterically couple dsDNA binding to cleavage. Biochemical^{11,14,15} and structural^{16,17} experiments using the extensively studied *SpyCas9* have revealed that its function is governed by a sophisticated allosteric mechanism that transfers gRNA and dsDNA binding information from the Rec lobe to the distal catalytic sites. A dynamically driven allosteric signal spans the HNH domain of *SpyCas9*, enabled by the plasticity of the Rec lobe, which orchestrates the conformational activation required for DNA cleavage.^{14,18} Our prior work revealed a divergence in the timescales of allosteric motions in the *SpyCas9* and *GeoCas9* HNH domains,^{7,17} suggesting an unusually flexible HNH and unique allosteric mode of regulation for *GeoCas9*. It is therefore also possible that docking of the gRNA with *GeoCas9*, and thus its interaction with the RNA:DNA hybrid, may differ from the *SpyCas9* system, as *GeoCas9* contains a truncated Rec lobe with only two of the three canonical subdomains.

The high thermal stability and more compact size of *GeoCas9* (it is 281 residues shorter than *SpyCas9*) can be especially important for *in vivo* delivery applications, since promising viral vectors (*i.e.* adeno-associated virus, AAV) have cargo capacities of ~4.7kb,¹⁹ which prevents

SpyCas9-gRNA packaging into a single AAV vector but permits “all-in-one” delivery of *GeoCas9*-gRNA.¹² Until the very recent cryo-EM structures of *GeoCas9*,^{20,21} little was known about specific residues that influence its structure, gRNA binding, or function. Our recent NMR work with *SpyCas9* uncovered pathways of micro-millisecond timescale motions that propagate chemical information related to allostery and specificity through *SpyRec* and its RNA:DNA hybrid,^{16,17} prompting us to investigate this phenomenon in *GeoRec*.

An atomic-level structural understanding of specificity in large multi-domain protein- nucleic acid complexes like Cas9 is often difficult to address by NMR spectroscopy. Although dynamic ensembles in DNA repair enzymes have provided some insight,²² many efforts to improve Cas9 specificity and reduce off-target activity have relied on large mutational screens²³ or error-prone PCR²⁴, which are less intuitive. Inter-subunit allosteric communication between the catalytic HNH domain and the Rec lobe is critical to Cas9 specificity, as the binding of off-target DNA sequences at Rec alter HNH dynamics to affect DNA cleavage.^{8,25,26} To further probe the fundamental role of protein motions in the function and specificity of *GeoCas9*, as well as the effect of protein-nucleic acid interactions on its structural signatures, we engineered two mutations in *GeoRec* (K267E and R332A, housed within *GeoRec2*). We hypothesized that these variants could enhance *GeoCas9* specificity (*i.e.* limit its off-target cleavage) for two reasons. First, the chosen mutation sites are homologous to those of specificity-enhancing variants of *SpyCas9*.^{24,27} Second, altered Cas9-gRNA interactions have been shown to be a consequence of specificity- enhancement and these charged residues appear to directly interact with the gRNA.^{10,11,15,28} Balancing these two points is the fact that Type-II Cas systems generally have conserved nuclease domains, but are delineated by highly varied Rec lobes.¹² This implies that the structural and dynamic properties of Rec may play an outsized role in differentiating the functions of *SpyCas9* and *GeoCas9*, which may not be identical. Nevertheless, our work provides new insight into the biophysical, biochemical, and functional role of the *GeoRec* lobe and how mutations modulate the domain itself and its interaction with gRNA in full-length *GeoCas9*.

Results

The structural similarity of *GeoRec1*, *GeoRec2*, and *GeoRec* facilitates NMR analysis of protein dynamics and RNA affinity

GeoCas9 is a 1087 amino acid polypeptide, thus we employed a “divide and concur” approach for NMR studies, which we previously showed to be useful for quantifying allosteric structure and motion in *SpyCas9*.^{16,17,29,30} The *GeoRec* lobe is comprised of subdomains *GeoRec1* and *GeoRec2*, which likely work together to recognize nucleic acids. We engineered constructs of the *GeoRec1* (136 residues, 16 kDa) and *GeoRec2* (212 residues, 25 kDa) subdomains and solved the X-ray crystal structure of *GeoRec2* at 1.49 Å, which aligns remarkably well with the structure of the *GeoRec2* domain within the AlphaFold model (RMSD 1.03 Å) and new cryo-EM structure of *GeoCas9* (RMSD 1.10 Å, **Figure 1A**). We were neither able to crystallize *GeoRec1* nor full-length *GeoCas9* in the apo state, but our *GeoRec2* crystal structure represents the structure of the subdomain within the full-length *GeoCas9* protein quite well. Our previous studies of *GeoHNH* also show identical superpositions of X-ray crystal structures with full-length Cas complexes.⁷ In addition to the individual subdomains, we also generated an NMR construct of the intact *GeoRec* (370 residues, 43 kDa).

Despite only 22% sequence identity, the structure of *SpyRec3* and *GeoRec2* are highly similar (RMSD 2.00 Å, **Figure S1**). The structure of *GeoRec1*, in contrast, does not align perfectly with *SpyRec1*, instead, it partially aligns with both *SpyRec1* and *SpyRec2* (**Figure S1**). Thus, the nearly identical *SpyRec3* and *GeoRec2* architectures and their intrinsic dynamics may be a common thread among Type II Cas9s of different size and PAM preference. To capture atomic-level signatures of *GeoRec*, we obtained well-resolved ¹H-¹⁵N NMR fingerprint spectra for all three

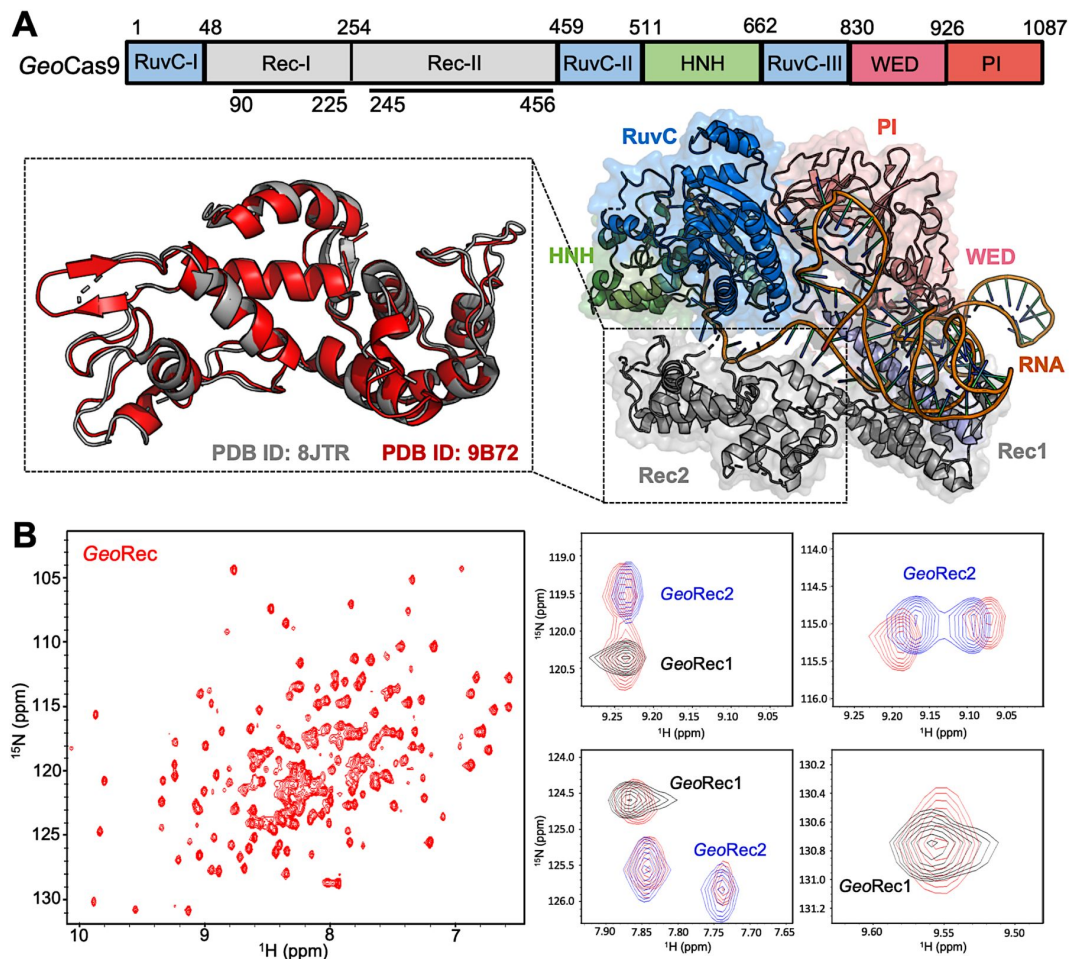


Figure 1.

(A) Arrangement of *GeoCas9* domains across the primary sequence. The cryo-EM structure of *GeoCas9* in complex with gRNA (PDB: 8JTR) shows poor resolution of HNH. The *GeoRec2* domain from PDB: 8JTR (gray) is overlaid with our X-ray structure of *GeoRec2* (red, PDB: 9B72). **(B)** $^1\text{H}/^{15}\text{N}$ TROSY HSQC NMR spectrum of *GeoRec* collected at 850 MHz. Overlays of this spectrum with resonances from spectra of *GeoRec1* (black) and *GeoRec2* (blue) demonstrate a structural similarity between the isolated subdomains and intact *GeoRec*.

protein constructs and assigned the amide backbones (Figure S2). ^1H - ^{15}N amide and ^1H - $^{13}\text{CH}_3$ Ile, Leu, and Val (ILV)-methyl NMR spectra (Figure 1B, S3) of *GeoRec* overlay very well with those of its individual subdomains, suggesting that the linkage of subdomains within the full-length *GeoRec* polypeptide does not alter their individual folds. Consistent with this observation, circular dichroism (CD) thermal unfolding profiles of *GeoRec1* ($T_m \sim 34^\circ\text{C}$) and *GeoRec2* ($T_m = 61.50^\circ\text{C}$) are distinct and occur as separate events in the unfolding profile of *GeoRec* (Figure S4). The dumbbell shape of *GeoRec*, with its two globular subdomains connected by a short flexible linker, is a likely contributor to these biophysical properties.

Rationally designed *GeoRec2* mutants do not substantially impact the *GeoRec* structure

To understand how the structure and gRNA interactions of *GeoCas9* can be modulated at the level of *GeoRec*, we engineered two charge-altering point mutants in the *GeoRec2* subdomain, K267E and R332A. Based on the AlphaFold2 model of *GeoCas9*, both of these residues are $< 5\text{\AA}$ from the bound RNA:DNA hybrid and were predicted to interface with the nucleic acids directly (Figure 2A/B). A new experimental cryo-EM structure of *GeoCas9* confirmed the interaction between K267 and the gRNA, but does not report a $< 5\text{\AA}$ interaction of R332 with the gRNA.³¹ The rationale for our designed mutations was also that removal of positive charge would weaken the interactions between *GeoCas9* and the gRNA, affecting K_d via the electrostatics or dynamics of the *GeoRec* lobe. Studies of *SpyCas9* revealed that interaction of *SpyRec3* (analogous to *GeoRec2*) with its RNA:DNA hybrid triggers conformational rearrangements that allow the catalytic HNH domain to sample its active conformation.¹¹ Thus, *SpyRec3* acts as an allosteric effector that recognizes the RNA:DNA hybrid to activate HNH. Mismatches (*i.e.* off-target DNA sequences) in the target DNA generally prevent *SpyRec3* from undergoing the full extent of its required conformational rearrangements, leaving HNH in a “proofreading” state with its catalytic residues too far from the DNA cleavage site. Off-target DNA cleavage by Cas9 remains an area of intense study and substantial effort from various groups has gone into mitigating such effects.^{15,23,28,32,33} Indeed, many high-specificity *SpyCas9* variants contain mutations within *SpyRec3* that increase the threshold for its conformational activation, reducing the propensity for HNH to sample its active state in the presence of off-target DNA sequences.^{11,15,23} Studies of flexibility within Rec itself, as well as its gRNA interactions in the presence of mutations, are therefore essential to connecting biophysical properties to function and specificity in related Cas9s. The K267E *GeoRec2* variant is sequentially and structurally similar to a specificity enhancing site in *SpyCas9* (K526E), within the *evoCas9* system.²⁷ The *SpyCas9* K526E mutation substantially reduced off-target activity alone, but was even more effective in conjunction with three other single-point mutations in *SpyRec3*.²⁷ The R332A *GeoRec2* variant also resembles one mutation within a high-specificity *SpyCas9* variant, an early iteration of HiFi *SpyCas9* called HiFi Cas9- R691A.²⁴ We assessed mutation-induced changes to local structure in *GeoRec* via NMR chemical shift perturbations in ^1H - ^{15}N HSQC backbone amide spectra. Consistent with experiments using *GeoRec2* alone, chemical shift perturbations and line broadening are highly localized to the mutation sites. (Figure 2C-F). Perturbation profiles of the *GeoRec2* subdomain and intact *GeoRec* also implicate the same residues as sensitive to the mutations (Figure S5).

CD spectroscopy revealed that wild-type (WT), K267E, and R332A *GeoRec2* maintained similar alpha-helical secondary structure, though the thermostability of both variants was slightly reduced from that of WT *GeoRec2* (Figure S6). The T_m of WT *GeoRec2* is $\sim 62^\circ\text{C}$, consistent with the T_m of the full-length *GeoCas9*, while that of K267E *GeoRec2* was decreased to $\sim 55^\circ\text{C}$. Though the R332A *GeoRec2* T_m remains $\sim 62^\circ\text{C}$, this variant underwent a smaller unfolding event near 40°C before completely unfolding. These data suggest that despite small structural perturbations, both mutations are destabilizing to *GeoRec2*, which led us to expect a change in NMR-detectable protein dynamics.

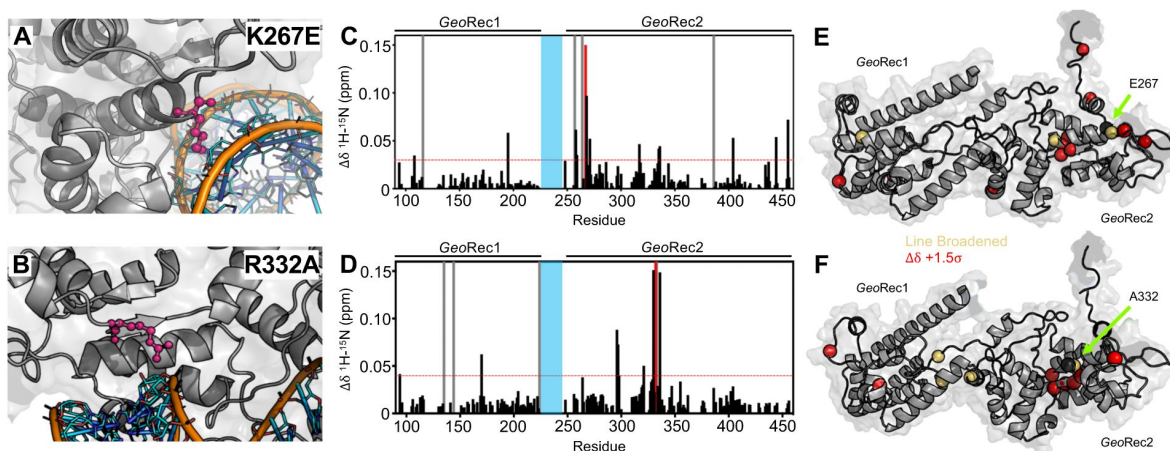


Figure 2.

(A, B) Sites of selected mutations within *GeoRec2*, K267 and R332, are highlighted as purple sticks directly facing the RNA and DNA modeled from *NmeCas9* (PDB ID: 6JDV), allowing for prediction of the binding orientation within *GeoCas9*. NMR chemical shift perturbations caused by the K267E (C) or R332A (D) mutations are plotted for each residue of *GeoRec*. Gray bars denote sites of line broadening, the blue bar denotes an unassigned region of *GeoRec* corresponding to the native Rec1-Rec2 linker, and the red bar indicates the mutation site. The red dashed line indicates 1.5σ above the 10% trimmed mean of the data. Chemical shift perturbations 1.5σ above the 10% trimmed mean are mapped onto K267E (E) and R332A (F) *GeoRec* (red spheres). Resonances that have broadened beyond detection are mapped as yellow spheres and the mutation sites are indicated by a black sphere and green arrow.

Mutations enhance and redistribute molecular motions within GeoRec2

Due to the high molecular weight of the intact *GeoRec* lobe, decays in NMR signal associated with spin relaxation experiments were significant and hampered data quality. Thus, we focused on quantifying the molecular motions of the *GeoRec2* subdomain, where the K267E and R332A mutations reside, and the chemical shift perturbations are most apparent. To obtain high-quality per-residue information representative of *GeoRec*, we measured longitudinal (R_1) and transverse (R_2) relaxation rates and heteronuclear ^1H - ^{15}N NOEs (Figure S7), then used these data in a Model-free analysis of per-residue order parameters (S^2). Previous measurements of S^2 across the adjacent *GeoHNN* nuclease revealed substantial ps-ns timescale flexibility,⁷ leading us to wonder whether a similar observation would be made for *GeoRec2*, which abuts *GeoHNN*. Such a finding could suggest that HNN-Rec2 crosstalk in *GeoCas9* is driven primarily by rapid bond vector fluctuations. However, unlike *GeoHNN*, S^2 values for *GeoRec2* are globally elevated, suggesting that the ps-ns motions of this subdomain arise primarily from global tumbling of the protein in solution. We therefore carried out Carr-Purcell-Meiboom-Gill (CPMG) relaxation dispersion NMR experiments to assess the flexibility of *GeoRec2* on slower timescales, which has been linked to chemical information transfer in the well-studied *SpyCas9*.^{10,16,17,29} Evidence of μs -ms motions (*i.e.* curved relaxation dispersion profiles) is observed in 17 residues within the *GeoRec2* core, spanning its interfaces to Rec1 and HNN (Figure 3B). Such motions are completely absent from *GeoHNN*, thus two neighboring domains, *GeoRec2* and *GeoHNN*, diverge in their intrinsic flexibility (at least in isolation), raising questions about the functional implications of these motions in *GeoRec2*. We previously showed that heightened flexibility of *SpyRec3* via specificity-enhancing mutations concomitantly narrowed the conformational space sampled by *SpyHNN*, highlighting a “motional trade-off” between the domains. Manipulation of the flexibility of *SpyCas9* and *GeoCas9* domains by mutagenesis also impacts aspects of nucleic acid binding and cleavage,^{1,10,14,15,29} which led us to investigate similar perturbations in *GeoRec2*.

Since *SpyRec3* and *GeoRec2* have similar structures and μs -ms flexibility, we speculated that charge-altering mutations would modulate the biophysical properties of *GeoRec* and the function of *GeoCas9*, as observed for *SpyCas9*. We investigated K267E and R332A *GeoRec2* with NMR spin relaxation, as described for WT *GeoRec2* (*vide supra*). An analysis of chemical exchange rates, k_{ex} , derived from dual-field CPMG relaxation dispersion show a global k_{ex} for WT *GeoRec2* of $147 \pm 41 \text{ s}^{-1}$.¹ The K267E mutation, which directly contacts the nucleic acids, shifts the globally fitted k_{ex} to $376 \pm 89 \text{ s}^{-1}$, while the R332A variant maintains a global k_{ex} similar to that of WT *GeoRec2* ($142 \pm 28 \text{ s}^{-1}$) and consistent with its similar thermal stability. The global fit of the K267E variant is based on CPMG profiles of 33 residues, while that of R332A is derived from 18 residues (Table S1). Interestingly, the residues participating in the global motions of both variants are distinct from those of WT *GeoRec2*, demonstrating that residue-specific flexibility is redistributed throughout *GeoRec2*, which suggests an altered intradomain molecular crosstalk within the larger *GeoRec*. Indeed, perturbation to NMR-detectable motions in *SpyCas9* rewired its allosteric signaling and enzymatic function.^{16,29} A similar dynamic modulation of *GeoCas9* may fine-tune its DNA cleavage, which has been demonstrated within the *GeoHNN* nuclease¹ and wedge (WED) domains.³² We also assessed the ps-ns fluctuations of *GeoRec2* variants (a negligible contribution to the WT *GeoRec2* dynamic profile) and calculated order parameters from R_1 , R_2 and ^1H - ^{15}N NOE relaxation measurements (Figure 3C). Bond vector fluctuations on the ps-ns timescale are only locally altered, thus the mutation-induced reshuffling of these motions is negligible ($\langle \Delta S^2 \rangle \leq 0.1$) and suggests that, like WT *GeoRec2*, ps-ns motion arises primarily from global tumbling in solution.

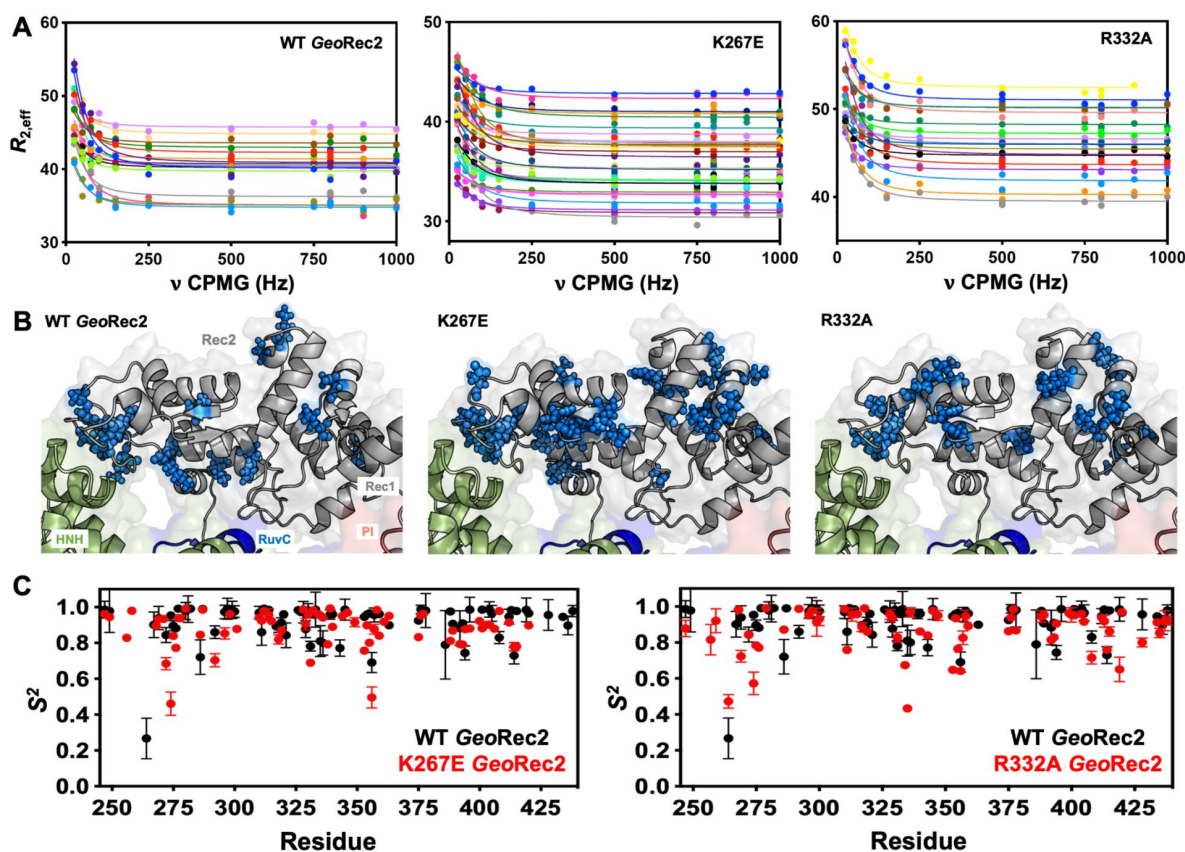


Figure 3.

(A) CPMG relaxation dispersion profiles of all residues with evidence of μs -ms motion, fit to a global k_{ex} of $147 \pm 41 \text{ s}^{-1}$ (WT GeoRec2, left), $376 \pm 89 \text{ s}^{-1}$ (K267E GeoRec2, center), and $142 \pm 28 \text{ s}^{-1}$ (R332A GeoRec2, right) collected at 25 °C and 600 MHz. Residues are colored in accordance with Table S1. Relaxation dispersion profiles for individual resonances are shown in Figures S8-S10. (B) Sites exhibiting CPMG relaxation dispersion in (A) are mapped to GeoRec as blue spheres. Adjacent domains within the cryo-EM structure of GeoCas9 are also shown. (C) Per-residue NMR order parameters of WT (black), K267E, and R332A (red, separate plots) GeoRec.

Mutations within GeoRec alter its affinity for RNA

The role of the Rec lobe in orienting the RNA:DNA hybrid within Cas9 is crucial to its function.^{10–13} Thus, the structure, motions, and nucleic acid interactions of Rec represent a critical piece of the Cas9 signaling machinery. Previous studies of *SpyCas9* revealed that gRNA binding to the Rec lobe induces a global structural rearrangement of the protein that positions the adjacent HNH into its “proofreading” state,¹⁵ after which target DNA binding positions the nucleases into active conformations for cleavage.^{15,34} We wondered if the atomistic details of the apo *GeoCas9*-to-RNP transition could be captured by NMR using the *GeoRec* construct. In our previous studies, we used an *in vitro* DNA cleavage assay with *GeoCas9* and a 141nt gRNA containing a 21nt spacer targeting the mouse *Tnnt2* gene locus.¹ Since this assay was already established, we utilized the same gRNA sequence. However, truncating this gRNA was necessary to optimize binding studies for NMR analysis. We focused on the 5' end of the gRNA, which includes the spacer sequence, based on the *GeoCas9* AlphaFold2 model and structural data from *NmeCas9* and *SpyCas9* showing interactions between the Rec lobe and this region of the gRNA. The subsequent cryo-EM structure of *GeoCas9* corroborated this interaction.²⁰ Initial attempts using a truncated 101nt portion of the full gRNA resulted in poor NMR spectra. An overlay of the ¹H-¹⁵N HSQC NMR spectra of apo *GeoRec* and *GeoRec*-RNP at a 1:1 molar ratio showed extensive line broadening (Figure S11), likely due to the large size of the complex (75.5 kDa). To mitigate this issue, a 39nt RNA containing the 21bp spacer sequence was selected for its ability to maintain the NMR signal while being long enough to interact fully with the Rec lobe, as suggested by prior structures (Figure S12). When bound to *GeoRec*, this complex is 55.6 kDa and a ¹H-¹⁵N NMR spectral overlay of apo *GeoRec* and the domain bound to 39nt RNA shows clear, resolved resonances with significant chemical shift perturbations and line broadening (Figure 4A/B, S11). The strongest chemical shift perturbations are localized to the *GeoRec2* subdomain that interfaces with the RNA:DNA hybrid at the PAM distal end, where previous studies of specificity-enhancing variants of *SpyCas9* have identified alterations in nucleic acid binding to *SpyRec3*.¹⁶ It is not known whether specific residues at the PAM distal binding interface of *GeoRec2* play a similar role. Line broadening is evident in both *GeoRec1* and *GeoRec2*, primarily localized to the RNA:DNA hybrid interface revealed in recent *GeoCas9* structures. Microscale thermophoresis (MST) experiments quantified the affinity of *GeoRec* for this RNA, producing a $K_d = 3.3 \pm 1.5 \mu\text{M}$ that is consistent with the concentration-dependent NMR chemical shift perturbations (Figure 5A).

To understand how the K267E and R332A mutants impact RNA binding to *GeoRec*, we conducted RNA titration experiments via NMR and observed that chemical shift perturbations were attenuated in both variants, relative to WT *GeoRec*. Despite this muted structural effect, the impact from RNA-induced line broadening remains substantial in the *GeoRec1* subdomain. Our NMR data revealed that a three-fold greater concentration of RNA was required to induce the maximal structural and dynamic effects in the variants than is required for WT *GeoRec* (Figure 4A/C), suggesting that the variants have a reduced RNA affinity. MST experiments showed statistically significant reductions in RNA affinity for the K267E and R332A constructs, relative to WT *GeoRec*, where K267E *GeoRec* produced a $K_d = 7.2 \pm 1.0 \mu\text{M}$ and R332A *GeoRec* produced a $K_d = 7.2 \pm 1.5 \mu\text{M}$ (Figure 5B). The ~2-fold increase in K_d may also be due, in part, to a change in the binding mode of the RNA, such as a faster k_{off} . Collectively, these data reveal that mutations within *GeoRec* primarily alter its structure around the mutation site with weaker distal effects, but more significantly impact protein dynamics and in turn, the RNA interaction. NMR experiments also demonstrate that the presence of RNA impacts both subdomains of *GeoRec*, providing a significant structural interface for additional molecular tuning of nucleic acid binding. To investigate the impact of RNA binding on *GeoRec2* in greater detail, we conducted NMR titration experiments using the isolated domain, which yielded even clearer NMR spectra. Figure S13 shows NMR spectra of WT, K267E, and R332A *GeoRec2* overlaid with their corresponding RNA-bound spectra (39nt *Tnnt2* RNA). At protein:RNA molar ratios used for full-length *GeoRec* studies, the WT *GeoRec2* spectrum exhibited significant line broadening across the *GeoRec2* sequence. Plots of

Figure 4.

(A) NMR chemical shift perturbations caused by RNA binding to WT, K267E, and R332A *GeoRec*. Gray bars denote sites of line broadening, and the blue bar denotes an unassigned region of *GeoRec* corresponding to the flexible Rec1-Rec2 linker. The red dashed line indicates 1.5σ above the 10% trimmed mean of the data. (B) Representative NMR resonance shifts caused by titration of a 39nt portion of the full gRNA into WT *GeoRec*. (C) NMR titration of 39nt RNA into K267E (top) and R332A (bottom) *GeoRec*. The left panel of each pair demonstrates that minimal change in NMR chemical shift or resonance intensity is apparent at RNA concentrations mimicking the WT titration. The right panel of each pair depicts the titration over a three-fold wider concentration range of RNA, where shifts and line broadening are visible. Representative resonances are colored by increasing RNA concentration in the legend.

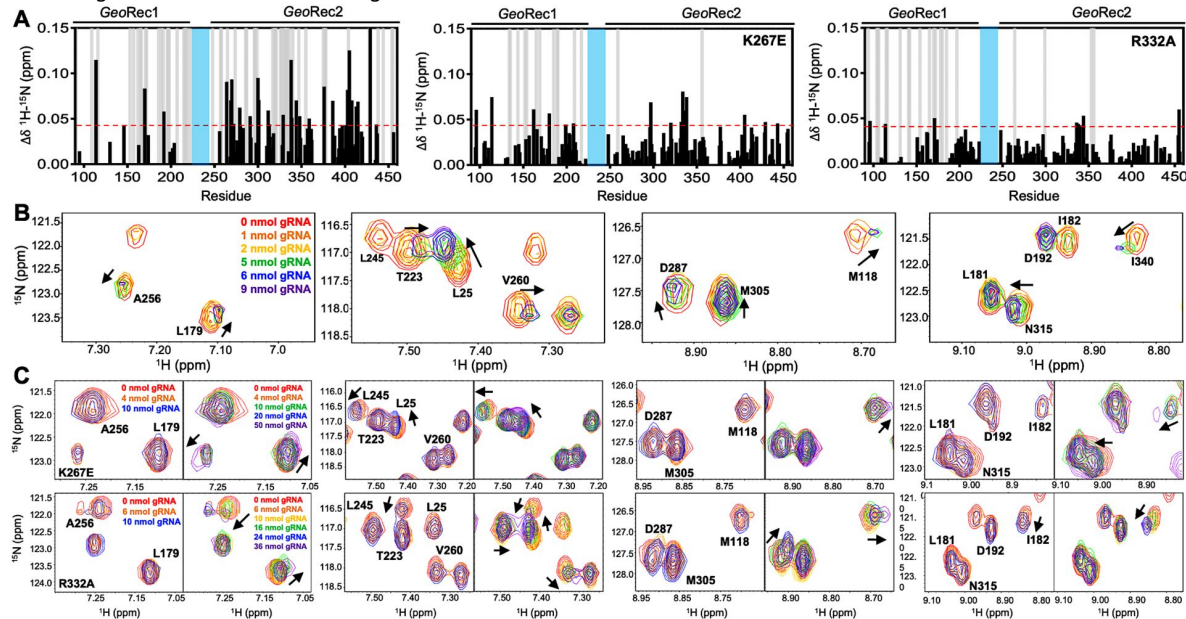
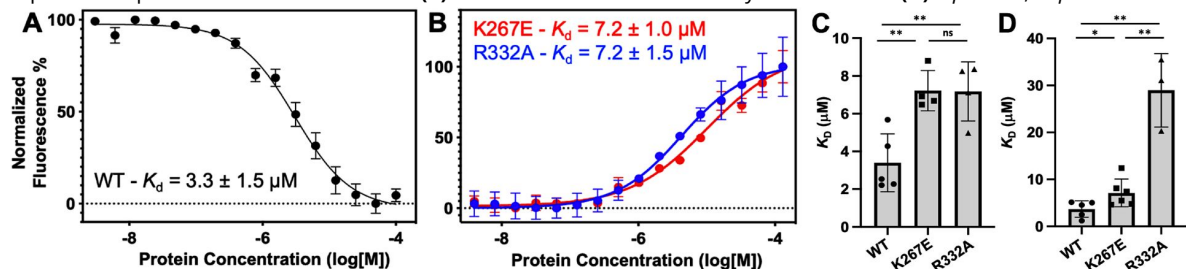


Figure 5.

Representative MST-derived profiles of WT (A), K267E, and R332A (B) *GeoRec* binding to a Cy5-labeled 39nt portion of a gRNA, yielding $K_d = 3.3 \pm 1.5 \mu\text{M}$, $K_d = 7.2 \pm 1.0 \mu\text{M}$ and $K_d = 7.2 \pm 1.5 \mu\text{M}$, respectively. Bar graphs comparing K_d values across $n \geq 3$ replicate samples are shown for Tnnt2 RNA (C) and 8UZA RNA from a recent cryo-EM structure (D). * $p < 0.05$, ** $p < 0.004$



NMR peak intensities ($I_{\text{bound}}/I_{\text{free}}$) show substantial resonance intensity losses (**Figure S13**), with many residues likely in the intermediate exchange regime, in addition to the assumed changes in rotational correlation of the domain. In comparison, spectra of the RNA-bound K267E and R332A *GeoRec2* variants showed less pronounced signal decay at the same levels of titrant, retaining nearly double the $I_{\text{bound}}/I_{\text{free}}$ ratio across these spectra (**Figure S13**). These data are consistent with the results of NMR experiments with the 43 kDa *GeoRec*, supporting the premise that *GeoRec2* mutations weaken its interaction with RNA. The less crowded NMR spectrum of isolated *GeoRec2* facilitated the resolution of distinct structural features that explain the impact of the mutations on RNA binding (**Figure S9**). For example, residue I53 adopts a similar conformation in RNA-bound WT and K267E *GeoRec2* but assumes a different structural state in R332A. Conversely, residue R25 populates a WT-like structure in RNA-bound R332A *GeoRec2*, unlike K267E. Additionally, two resonances are observed for residue K71 in the RNA-bound R332A NMR spectrum, indicating real-time equilibration between two structural states. This effect is unique to the R332A variant and underscores subtle structural and dynamic changes to *GeoRec* during RNA binding.

We further examined the NMR data to attempt to identify residues most critical for RNA binding to *GeoRec*. In an overlay of the WT and mutant RNA-induced chemical shift perturbations ($\Delta\delta$, **Figure S14**), it became clear that the effect of RNA binding to *GeoRec* variants was muted, where even at saturating concentrations, the chemical shift perturbations across the K267E and R332A *GeoRec2* sequences were weaker than those of same residues in WT *GeoRec*. The residual $\Delta\delta$ (WT - mutant) was plotted (**Figure S14**), where positive values indicate that residues in a *GeoRec* variant are weakly affected by RNA, relative to WT. Negative residual $\Delta\delta$ denote sites where *GeoRec* variants experience a greater structural impact from RNA than corresponding sites in WT. Of particular interest are the positive residuals that hint at the sites in *GeoRec* most critical for tight RNA binding. These residues were mapped onto the *GeoRec* structure (**Figure S14**) and termed allosteric hotspots, as many are not at the RNA interface. Mutations of these hotspots in future studies offers a potential means of precisely tuning the affinity of *GeoRec* to its gRNA. Notably, residues with positive residual $\Delta\delta$ (suggested as critical for tight RNA binding) largely overlap in the analysis of both variants. Specifically, residues F170, R192, H264, R269, L270, L279, H300, D301, E368, D376, D403, E405, E408, and I429 appear as allosteric hotspots (with CPMG relaxation dispersion) critical to WT-like RNA interaction.

Having observed a reduced affinity of *GeoRec* variants for RNA by NMR and MST, we next quantified the impact of the K267E and R332A mutations on RNP formation and stability in full-length *GeoCas9*. The thermal unfolding midpoint of full-length WT *GeoCas9* determined by CD is $\sim 60^\circ\text{C}$ and the K267E and R332A mutations do not change the T_m of the apo protein (**Figure S15**). Upon formation of an RNP (using a full-length gRNA), the T_m of WT *GeoCas9* increases to 73°C . K267E *GeoCas9* retains a similar T_m increase to 70°C , while R332A *GeoCas9* forms a less stable RNP with T_m of 61°C . The trend of these data is consistent with NMR and MST, which highlight that although K267E and R332A mutations within *GeoRec* have somewhat muted structural effects, these changes alter protein dynamics and the interaction with gRNA.

Mutations in full-length *GeoCas9* alter its structural dynamics and interaction with gRNA

To further investigate the effects of mutations on protein dynamics, we performed molecular dynamics (MD) simulations based on the cryo-EM structure of full-length *GeoCas9* (PDB: 8UZA) in complex with gRNA and target DNA. We simulated the full-length WT *GeoCas9* and its K267E and R332A mutants as well as a double mutant combining K267E and R332A (**Figure 6A**), in three replicates of approximately 2 μs each. Multi-microsecond simulations revealed substantial changes in the dynamics of the *GeoCas9* mutants compared to the WT (**Figure S16**). Specifically, we observed that mutations in the REC domain significantly altered the dynamics of both the Rec and the adjacent HNH (**Figure S16**). Differential root-mean-square fluctuations (ΔRMSF)

analysis of protein residues between the WT and variants further highlighted these alterations, showing increased dynamics in the HNH and Rec domains induced by the mutations (**Figure 6B**). To quantify the impact of these mutations, we analyzed protein- RNA interactions by calculating the number of contacts between *GeoCas9* and gRNA in the WT and mutant systems. A contact was defined as a distance between two atoms of ≤ 4.5 Å. The number of contacts was significantly reduced in all variants compared to the WT, with the most pronounced reduction observed in the K267E variant (**Figure 6C**). We next quantified gRNA-Rec domain binding by calculating the binding free energy using the MM-GBSA method over ~ 200 ns of stable simulation trajectories (details in Materials and Methods). Consistent with a reduction in gRNA contacts, variants with the K267E mutation exhibited a substantial reduction in binding free energy (>60 kcal mol⁻¹) relative to the WT, whereas the R332A variant displayed a smaller reduction (<20 kcal mol⁻¹, **Figure 6D**). Notably, protein-DNA interactions remained largely unaffected, suggesting that these mutations do not impair *GeoCas9* DNA cleavage ability.

Additionally, we simulated a novel variant, i*GeoCas9* (PDB: 8UZZ), containing mutations in the Rec1 and WED domains (**Figure 6A**, mutations highlighted in lime green). This variant was recently demonstrated to have enhanced specificity in genome-editing.²¹ Intriguingly, i*GeoCas9* exhibited increased dynamics in the HNH and Rec domains, along with a reduction in gRNA binding free energy similar to the K267E and R332A mutants. These results suggest a distinct allosteric pathway involving additional residues that enables i*GeoCas9* to maintain improved DNA cleavage activity despite reduced gRNA binding affinity. In fact, i*GeoCas9* samples the greatest conformational space of any variant tested (**Figure S16**), suggesting a high level of flexibility is critical to enhanced specificity in *GeoCas9*. Collectively, our simulations reveal that mutations K267E and R332A destabilize the *GeoCas9* interaction with gRNA, consistent with NMR observations. Furthermore, the enhanced dynamics and altered binding affinities observed in i*GeoCas9* indicate potential allosteric mechanisms that optimize its genome-editing functionality, where K267E and R332A evoke a similar, but lesser degree of biophysical change.

DNA cleavage assays suggest the highly stable *GeoCas9* is resistant to functional changes by K267E or R332A mutations

The dynamic impact of the *GeoRec* mutations and their altered gRNA interactions at the biophysical level led us to speculate that either mutation incorporated into full-length *GeoCas9* would also alter its DNA cleavage function, especially at elevated temperatures where WT *GeoCas9* is most active. Temperature-dependent functional alterations were previously observed for single-point mutations within *GeoHNH*.¹ Although the K267E and R332A mutations slightly diminished on-target DNA cleavage by *GeoCas9*, the effect was very subtle and these overall cleavage activities followed the temperature dependence of WT *GeoCas9* quite closely (**Figure S17**).

To assess the impact of the K267E and R332A mutations on *GeoCas9* specificity, we assayed the propensity for off-target cleavage using DNA substrates with mismatches 5-6 or 19- 20 base pairs from the PAM site (**Figure S18**, Table S2). As a control for on- and off-target activity, we assayed WT *SpyCas9* alongside the widely used high-specificity HiFi-*SpyCas9* variant²⁴ (**Figure S18**, Table S3) and found a lower percent of digested off-target (mismatched) DNA sequences when compared to WT *SpyCas9*. As expected, WT *GeoCas9* was increasingly sensitive to mismatched target sequences closer to the seed site, which has been demonstrated with *SpyCas9* and other Cas systems.^{2,6,8,35} No significant differences in activity were observed with digestion durations ranging from 1-60 minutes,² implying that a 1-minute digestion is sufficient for *in vitro* activity of *GeoCas9* with the target DNA template. While these findings generally align with prior investigations of off-target DNA cleavage,^{2,6,8} there are nuanced differences. Specifically, a previous study reported $\sim 10\%$ cleavage of off-target DNA with a mismatch 5-6 base pairs from the PAM by WT *GeoCas9*.² Our results showed nearly 50% cleavage for the same off-target mismatch, but still a significant decrease in cleavage from on-target or 19-20 base pair distal mismatches. This could be due to the relatively high RNP concentrations (600-900 nM) in our assay

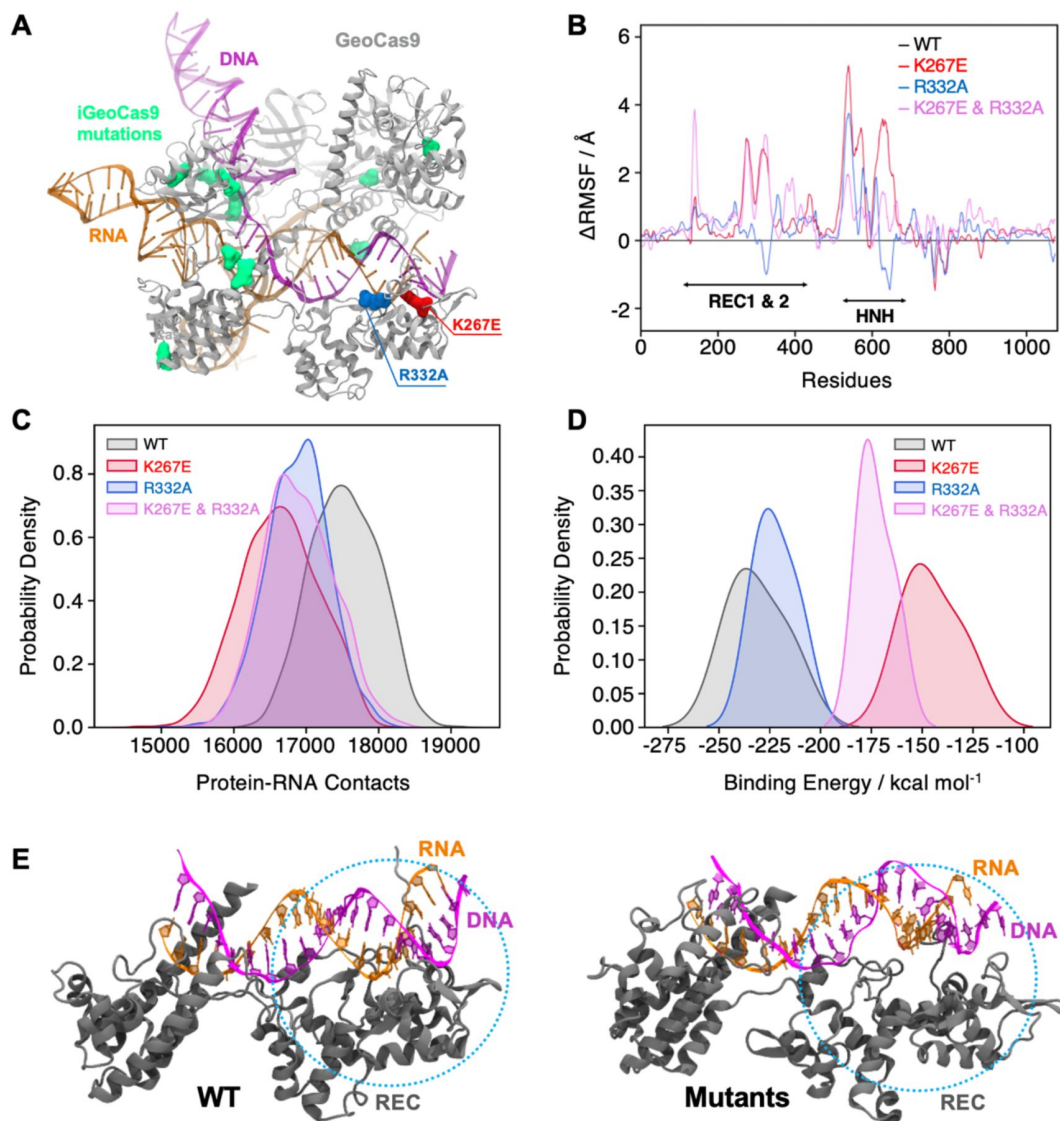


Figure 6.

Effects of mutations in full-length *GeoCas9* revealed by MD simulations. **(A)** The structure of *GeoCas9* (PDB: 8UZA, protein in gray) bound to gRNA (orange) and DNA (magenta) is shown. Mutations studied include K267E (red), R332A (blue), and the 10 mutations of i*GeoCas9* (lime green), all highlighted in surface representation. **(B)** Differential root-mean-square fluctuations (ΔRMSF) of protein residues computed between WT *GeoCas9* and the K267E (red), R332A (blue), and double mutant (pink). **(C)** Distribution of protein-RNA contacts for WT and *GeoCas9* variants computed over the 6 μs simulation ensemble. **(D)** Comparison of gRNA binding free energy to the Rec domain in WT *GeoCas9* and variants. **(E)** Representative snapshots from MD simulations illustrating structural changes in Rec-gRNA association in WT *GeoCas9* (left) and variants (right).

(for clear visibility on the gel), compared to prior studies with RNP concentrations ≤ 500 nM.² Our results corresponded closely to those of prior studies with a 19-20 base pair mismatch, where off-target cleavage is tolerated by WT *GeoCas9*.² Single-point mutants K267E and R332A *GeoCas9* have negligible impact on *GeoCas9* specificity (both variants follow the trend of WT *GeoCas9*, Figure S18), which contrasts prior work with *SpyCas9* that demonstrated robust specificity enhancement with single-point mutations in Rec.^{24,27} Additionally, a *GeoCas9* double mutant K267E/R332A exhibits decreased on-target cleavage efficiency, which has been noted in high-specificity Cas systems.^{15,23,24,27,36–38} However, the additive effect of the K267E/R332A double mutant still does not enhance *GeoCas9* specificity in our assay.

MST-derived binding affinities using full-length Tnnt2 gRNA and full-length WT, K267E, or R332A *GeoCas9* indicate that all three proteins have similar affinities for the gRNA used in the functional assays (Figure 7). Thus, mutations do not substantially alter full-length *GeoCas9* binding to Tnnt2 gRNA, supporting similar cleavage activities for these proteins. We repeated the MST experiments using a different gRNA sequence derived from the new cryo-EM structure of *GeoCas9* (PDB:8UZA), which has a different spacer sequence. We observed a similar trend, as all *GeoCas9* variants exhibited comparable affinities for this gRNA. Our functional studies illustrate an apparent resilience of *GeoCas9* to major functional changes at the level of Rec, despite comparable mutations having profound functional impacts in mesophilic Cas9s.

Discussion

CRISPR-Cas9 is a powerful tool for targeted genome editing with high efficiency and modular specificity.^{2,15,24,27} Allosteric signals propagate DNA binding information to the HNH and RuvC nuclease domains, facilitating their concerted cleavage of double-stranded DNA.^{8,10,14,15} The intrinsic flexibility of the nucleic acid recognition lobe plays a critical role in this information transfer, exerting a measure of conformational control over catalysis.¹⁰ This study provides new insights into the structural, dynamic, and functional role of the thermophilic *GeoCas9* recognition lobe. Novel constructs of subdomains *GeoRec1* and *GeoRec2*, as well as intact *GeoRec* show a high structural similarity to the domains in full-length *GeoCas9*, facilitating solution NMR experiments that captured the intrinsic allosteric motions across *GeoRec2*. These studies revealed the existence of μ s-ms timescale motions that are classically associated with allosteric signaling and enzyme function, which span the entire *GeoRec2* domain to its interfaces with *GeoRec1* and the adjacent *GeoHNH* domain.

Based on homology to specificity-enhancing variants of the better studied *SpyCas9*, the biophysical and biochemical consequences of two mutations were tested in *GeoRec2*, the larger *GeoRec* lobe, and full-length *GeoCas9*. We speculated that removing positively charged residues with potential to interact with negatively charged nucleic acids could disrupt *GeoCas9*-gRNA complex formation, stability, and subsequent function by altering the protein or nucleic acid motions. Indeed, CPMG relaxation dispersion experiments revealed that mutations enhanced and reorganized the μ s-ms flexibility of *GeoRec2*. Further, NMR titrations showed the affinity of K267E and R332A *GeoRec* for 39nt portions of two gRNAs to be weaker than that of WT *GeoRec*, consistent with MST-derived K_d values using the isolated domain. The mutations also diminished the stability of the full-length *GeoCas9* RNP complex, though this could result from an allosteric effect that destabilizes the *GeoCas9* structure without appreciably altering gRNA binding.

The collective changes to protein dynamics, gRNA binding, and RNP thermostability suggested that mutations could modulate *GeoCas9* function, as observed in similar studies of *SpyCas9* reporting that gRNA dynamics, affecting the potential for the RNA:DNA hybrid to dissociate, have affected function.^{11,15,39} Yet, the functional impact of single-point and double mutations in this work were negligible, despite homologous K-to-E and R-to-A single point mutations enhancing specificity of the mesophilic *SpyCas9*. The biophysical impact of mutations within *GeoRec2* and *GeoRec* may

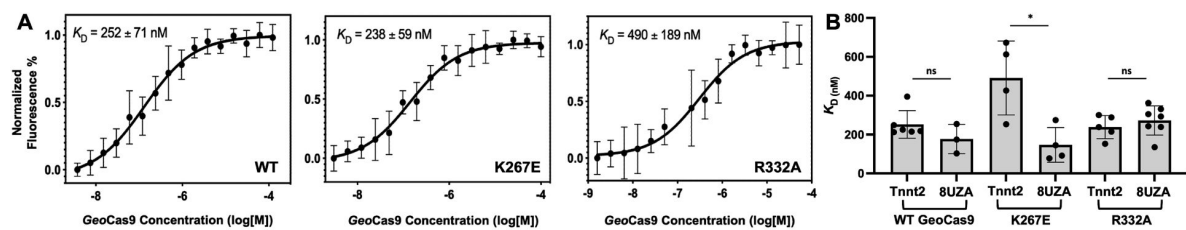


Figure 7.

(A) Representative MST-derived profiles of WT, K267E, and R332A GeoCas9 binding to a Cy5-labeled full-length Tnnt2 gRNA. (B) Bar graph comparing K_D values across $n \geq 3$ replicate samples are shown for Tnnt2 and 8UZA gRNA from a recent cryo-EM structure of GeoCas9. * $p < 0.01$

be tempered by its evolutionary resilience and the highly stable neighboring domains in the context of full-length *GeoCas9*, reflected in an unchanged affinity for target DNA once the RNP was formed.^{40,41} Thus, a greater number of additive (or synergistic) mutations within *GeoRec* would be required to fine-tune activity or specificity to a large degree.

It should be noted that the effects of these and other *GeoRec* mutations may vary *in vivo* or with alternative target cleavage sites and cell types. Such studies will be the subject of future work, as will biochemical assays of homologous mutations across diverse Cas9s, which have contributed to the wide use of CRISPR technology.⁴² We also note that despite the homology between *GeoRec2* and *SpyRec3* and the latter's role in evo- and HiFi-*SpyCas9* variants that inspired the K267E and R332A mutations, the maximally enhanced *SpyCas9* variants contain four mutations each. Presumably each individual substitution plays a small role modulating specificity. However, there is no consistent pattern that discerns whether multiple mutations will have additive or synergistic impacts on Cas9 function. NMR and MD studies of high-specificity *SpyCas9* variants (HF-1, Hypa, and Evo, each with distinct mutations in the *SpyRec3* domain) reveal universal structural and dynamic variations in regions of *SpyRec3* that interface with the RNA;DNA hybrid.¹⁶ Notably, a recently published variant, i*GeoCas9*,³² demonstrated enhanced genome-editing capabilities in HEK293T cells with eight mutations, though none in the Rec2 subdomain. This study highlighted the functional adaptability of i*GeoCas9* under low magnesium conditions, a trait beneficial in mammalian cells, distinguishing it from WT *GeoCas9*. These very recently published data, as well as the findings reported here still advance our molecular understanding of the functional handles in *GeoCas9*, relevant to the design of new enhanced variants.

This study marks the first phase of mapping allosteric motions and pathways of information flow in the *GeoRec* lobe with solution NMR experiments. Such information transfer is critical to the crosstalk between Rec and HNH in several Cas9s. Despite NMR advancements in perdeuteration,⁴³ transverse relaxation-optimized spectroscopy (TROSY),⁴⁴ and sparse isotopic labeling,⁴⁵ per-residue dynamics underlying allosteric signaling in large multi-domain proteins such as *GeoCas9* (~126 kDa) have remained challenging to characterize. Novel cryo-EM structures of *GeoCas9*³² will facilitate the merging of future NMR and MD simulation studies to report on RNP dynamics and atomic level networks of communication. The identification of additional (or synergistic) allosteric hotspots within *GeoRec* using an integrated workflow will help to further resolve the balance between structural flexibility and the unusually high stability of *GeoCas9*, leading to new insight into targeted manipulation of RNA affinity and enhanced variants.

Here, we set out to biophysically characterize the Rec lobe of *GeoCas9* to obtain new understanding of its function (in the context of well-studied mesophilic Cas9s). Using an AlphaFold2 model, and later a cryo-EM structure of *GeoCas9*, we introduced mutations based on proximal gRNA interactions and homology to specificity-enhancing sites in *SpyCas9*. However, the mutations did not affect *GeoCas9* function as expected, highlighting the complicated interplay between the biophysics of mesophilic and thermophilic Cas enzymes and the difficulty of applying universal functional predictions to Cas9. The very recent report of the i*GeoCas9* variant further reinforces this point.²¹ While high-specificity *SpyCas9* variants are heavily mutated in Rec3 (analogous to *GeoRec2*), i*GeoCas9* lacks mutations in Rec2 entirely, raising new questions about the functional role of *GeoRec*. MD simulations of WT *GeoCas9*, i*GeoCas9*, and the Rec variants revealed that while K267E and R332A induce dynamic effects on a similar trajectory to i*GeoCas9*, a true high-specificity variant samples a very wide conformational space with displacements of both Rec and HNH (**Figures 6** and **S16**). Through further study of the fundamental mechanism of *GeoCas9*, it remains possible that engineering of *GeoRec* may produce high-specificity variants.

Materials and methods

Expression and purification of *GeoRec1*, *GeoRec2*, *GeoRec*, and *GeoCas9*

The *GeoRec1* (residues 90–225) and *GeoRec2* (residues 245–456) subdomains, as well as the entire *GeoRec* lobe (residues 90–456) of *G. stearothermophilus* Cas9 were engineered into a pET28a vector with a N-terminal His₆-tag and a TEV protease cleavage site. The K267E and R332A mutations were separately introduced into the *GeoRec2* plasmid. Plasmids were transformed into BL21 (DE3) cells (New England Biolabs). Protein samples for CD spectroscopy, MST, and functional assays were grown in Lysogeny Broth (LB, Fisher), while isotopically labeled samples for NMR were grown in M9 minimal media (deuterated for *GeoRec2* and *GeoRec*) containing CaCl₂, MgSO₄, MEM vitamins, and 1.0 g/L ¹⁵N ammonium chloride and 2.0 g/L ¹³C glucose (Cambridge Isotope Laboratories), as the sole nitrogen and carbon sources, respectively. Cells were induced with 1 mM IPTG after reaching an OD₆₀₀ of 0.8–1.0 and grown for 4 hours at 37 °C post induction. The cells were harvested by centrifugation, resuspended in a buffer of 50 mM Tris-HCl, 250 mM NaCl, 5 mM imidazole, and 1 mM PMSF at pH 7.4, lysed by ultrasonication, and purified by Ni-NTA affinity chromatography. Following TEV proteolysis of the terminal His-tag, the samples were further purified on a Superdex75 size exclusion column. NMR samples were dialyzed into a buffer containing 20 mM NaPi, 80mM KCl, 1mM DTT, and 1mM EDTA at pH 7.4.

The full-length *GeoCas9* plasmid was acquired from Addgene (#87700), expressed in TB media and was expressed and purified as previously described.² The K267E, R332A, and K267E/R332A variants were introduced into full-length *GeoCas9* by modifying the original plasmid acquired from Addgene.

NMR spectroscopy

Backbone resonance assignments of *GeoRec1* and *GeoRec2* were carried out on a Bruker Avance NEO 600 MHz spectrometer at 25 °C. The following triple resonance experiments were collected for each sample: ¹H-¹⁵N TROSY-HSQC, HNCA, HN(CO)CA, HN(CA)CB, HN(COCA)CB, HN(CA)CO and HNCO. All spectra were processed in NMRPipe⁴⁶ and analyzed in Sparky⁴⁷. Three-dimensional correlations and assignments were made in CARRA⁴⁸ and *GeoRec1* and *GeoRec2* backbone assignments were deposited in the BMRB under accession numbers 52363 and 51197, respectively. Backbone resonance assignments of *GeoRec* were completed by transferring assignments from the individually assigned spectra of *GeoRec1* and *GeoRec2*, as done previously for other large Cas9 fragments.^{49,50}

NMR spin relaxation experiments were carried out in a temperature-compensated manner at 600 and 850 MHz on Bruker Avance NEO and Avance III HD spectrometers, respectively. CPMG experiments were adapted from the report of Palmer and coworkers⁵¹ with a constant relaxation period of 20 ms and ν_{CPMG} values of 0, 25, 50, 75, 100, 150, 250, 500, 750, 800, 900, and 1000 Hz. Exchange parameters were obtained from global fits of the data carried out with RELAX⁵² using the R2eff, NoRex, and CR72 models, as well as in-house fitting in GraphPad Prism with the following models:

Model 1: No exchange

$$R_2^{eff} = R_2^0 \quad (1)$$

Model 2: Two-state, fast exchange (Meiboom equation ⁵³)

$$R_2^{eff} = R_2^0 + \frac{\phi}{k_{ex}} \left[1 - \frac{4\nu_{CPMG}}{k_{ex}} \tanh \left(\frac{k_{ex}}{4\nu_{CPMG}} \right) \right] \quad (2)$$

Global fitting of CPMG profiles was determined to be superior to individual fits based on the Akaike Information Criterion.⁵⁴ Uncertainties in these rates were determined from replicate spectra with duplicate relaxation delays of 0, 25, 50 (×2), 75, 100, 150, 250, 500 (×2), 750, 800 (×2), 900, and 1000 Hz.

Longitudinal and transverse relaxation rates were measured with randomized T_1 delays of 0, 20, 60, 100, 200, 600, 800, and 1200 ms and T_2 delays of 0, 16.9, 33.9, 50.9, 67.8, 84.8, and 101.8 ms. Peak intensities were quantified in Sparky and the resulting decay profiles were analyzed in Sparky with errors determined from the fitted parameters. Uncertainties in these rates were determined from replicate spectra with duplicate relaxation delays of 20 (×2), 60 (×2), 100, 200, 600 (×2), 800, and 1200 ms for T_1 and 16.9, 33.9 (×2), 50.9 (×2), 67.8 (×2), 84.8, 101.8 (×2) ms for T_2 . Steady-state ^1H - ^{15}N NOE were measured with a 6 second relaxation delay followed by a 3 second saturation (delay) for the saturated (unsaturated) experiments and calculated by $I_{\text{sat}}/I_{\text{ref}}$. All relaxation experiments were carried out in a temperature-compensated interleaved manner.

Model-free analysis was carried out by fitting relaxation rates to five different forms of the spectral density function with local τ_m , spherical, prolate spheroid, oblate spheroid, or ellipsoid diffusion tensors.^{55–60} The criteria for inclusion of resonances in the diffusion tensor estimate was based on the method of Bax and coworkers.⁶¹ N-H bond lengths were assumed to be 1.02 Å and the ^{15}N chemical shift anisotropy tensor was -160 ppm. Diffusion tensor parameters were optimized simultaneously in RELAX under the full automated protocol.⁵² Model selection was iterated until tensor and order parameters did not deviate from the prior iteration.

NMR titrations were performed on a Bruker Avance NEO 600 MHz spectrometer at 25 °C by collecting a series of ^1H - ^{15}N TROSY HSQC spectra with increasing ligand (*i.e.* RNA) concentration. The ^1H and ^{15}N carrier frequencies were set to the water resonance and 120 ppm, respectively. Samples of WT, K267E, and R332A *GeoRec* were titrated with a 39nt portion of a gRNA until no further spectral perturbations were detected. NMR chemical shift perturbations were calculated as:

$$\Delta\delta = \sqrt{(\Delta\delta_{\text{HN}}^2 + \Delta\delta_{\text{NH}}^2/25)/2}$$

Microscale thermophoresis (MST)

MST experiments were performed on a Monolith X instrument (NanoTemper Technologies), quantifying WT, K267E, R332A, and K267E/R332A *GeoRec* binding to a 39-nt Cy5-labeled RNA at a concentration of 20 nM in a buffer containing 20 mM sodium phosphate, 150 mM KCl, 5 mM MgCl_2 , and 0.1% Triton X-100 at pH 7.6. The *GeoRec* proteins were serially diluted from a 200 μM stock into 16 microcentrifuge tubes and combined in a 1:1 molar ratio with serially diluted gRNA from a 40 nM stock. After incubation for 5 minutes at 37 °C in the dark, each sample was loaded into a capillary for measurement. K_d values for the various complexes were calculated using the MO Control software (NanoTemper Technologies). Statistical significance was calculated using a two-tailed T-test.

Circular dichroism (CD) spectroscopy

All *GeoCas9* and *GeoRec* proteins were buffer exchanged into a 20 mM sodium phosphate buffer at pH 7.5, diluted to 1 μM , and loaded into a 2 mm quartz cuvette (JASCO instruments). A CD spectrum was first measured between 200 - 250 nm, after which the sample was progressively heated from 20 – 90 °C in 1.0 °C increments while ellipticity was monitored at 222 and 208 nm.

Phosphate buffer baseline spectra were subtracted from the sample measurements. Prior to CD measurements, *GeoCas9*-RNP was formed by incubating 3 μM *GeoCas9* with its gRNA at a 1:1.5 molar ratio at 37 °C for 10 minutes. The unfolding CD data was fit in GraphPad Prism to:

$$\text{Ellipticity}(T) = \frac{[(m_f T + b_f) + (m_u T + b_u)] \exp\left[\left(-\frac{\Delta H_{D.VH}}{R}\right)\left(\frac{1}{T} - \frac{1}{T_m}\right)\right]}{1 + \exp\left[\left(-\frac{\Delta H_{D.VH}}{R}\right)\left(\frac{1}{T} - \frac{1}{T_m}\right)\right]}$$

X-ray crystallography

GeoRec protein purified as described above was crystallized by sitting drop vapor diffusion at room temperature by mixing 1.0 μL of 15 mg/mL *GeoRec* in a buffer of 20 mM HEPES and 100 mM KCl at pH 7.5 with 2.0 μL of crystallizing condition: 0.15 M calcium chloride, 15 % polyethylene glycol 6000, 0.1 M HEPES at pH 7.0. Crystals were cryoprotected in crystallizing condition supplemented with 30% ethylene glycol. Diffraction images were collected at the NSLS- II AMX beamline at Brookhaven National Laboratory under cryogenic conditions. Images were processed using XDS⁶² and Aimless in CCP4.⁶³ Chain A of the *N. meningitidis* Cas9 X-ray structure (residues 249-445 only, PDB ID: 6JDQ) was used for molecular replacement with Phaser followed by AutoBuild in Phenix.⁶⁴ Electron density was only observed for the *GeoRec2* subdomain. The *GeoRec2* structure was finalized through manual building in Coot⁶⁵ and refinement in Phenix.

Molecular dynamics (MD) simulations

Molecular Dynamics (MD) simulations were based on the cryo-EM structure of full-length *GeoCas9* (PDB: 8UZA, resolution 3.17 Å) in complex with gRNA and target DNA with two mutations in *GeoCas9* (at residues 8 and 582). Four systems were considered for the MD studies: WT, K267E, R332A, K267E/R332A and *iGeoCas9*. We generated the WT *GeoCas9* by back-mutating A8D and A582H from the cryo-EM structure (PDB: 8UZA), followed by introducing the mutations K267E, R332A, or a double mutation (with both K267E and R332A) for the variant systems. Subsequently, we performed MD simulation of *iGeoCas9* (PDB: 8UZH, resolution 2.63 Å) consisting of 10 mutations (D8A, E149G, T182I, N206D, P466Q, H582A, Q817R, E843K, E854G, K908R). All systems were solvated with explicit water in a periodic box of ~ 134 Å $\times$ ~ 154 Å $\times$ ~ 151 Å resulting in $\sim 276,000$ atoms. Counter ions were added to neutralize the systems. MD simulations were performed using a protocol tailored for protein-nucleic acid complexes,⁶⁶ previously applied in studies of CRISPR-Cas systems.^{67–69} All the simulations were performed by using Amber ff19SB force field for protein,⁷⁰ ff99bsc1 corrections and χOL3 corrections for DNA and RNA, respectively.^{71,72} Water molecules were described by TIP3P model.⁷³ All bonds involving hydrogens were constrained using the LINCS algorithm. A particle mesh Ewald method (PME) with a 10 Å cutoff was used to calculate electrostatics. Energy minimization was performed to relax the water molecules and counterions, keeping the protein-nucleic acid complex fixed with harmonic potential restraints of 100 kcal/mol Å².⁷⁴ Equilibration was performed by gradually increasing the temperature from 0 to 100 K and then to 200 K in canonical NVT ensemble and isothermal-isobaric NPT ensemble. A final temperature of 300 K was maintained via Langevin dynamics with a collision frequency $\gamma = 1/\text{ps}$ and a reference pressure of 1 atm was achieved through Berendsen barostat. Production runs were carried out in NVT ensemble for 2 μs for each system in three replicates, resulting in 6 μs per system (totaling 30 μs for all systems). The equations of motion were integrated with the leapfrog Verlet algorithm with a time step of 2 fs. All simulations were conducted using the GPU-empowered version of AMBER 22.⁷⁴ Analysis was performed on the aggregate ensemble (i.e., ~ 6 μs per system).

To characterize the protein-nucleic acid interactions in all the systems under investigation, we performed contact analysis. A contact was considered to form between two atoms within a cutoff distance of ≤ 4.5 Å. The binding free energy of gRNA and DNA with *GeoCas9* was calculated using the Molecular Mechanics Generalized Born Surface Area (MM-GBSA) method.^{75–77} This

approach was used to compare the Rec-gRNA binding affinity of WT *GeoCas9* with its mutants. For each system, the binding energies were calculated over the ~200 ns ensemble of the stable trajectories at an interval of ~20 ns.

DNA cleavage assays

GeoCas9 gRNA templates containing 21-nt spacers targeting the mouse *Tnnt2* gene locus were introduced into EcoRI and BamHI sites in pUC57 (Genscript). The plasmid was transformed into BL21(DE3) cells (New England BioLabs) and subsequent restriction digest of the plasmid DNA was carried out using the BamHI restriction enzyme (New England BioLabs) according to the manufacturer's instructions. Linearized plasmid DNA was immediately purified using the DNA Clean and Concentrator-5 kit (Zymo Research) according to the manufacturer's instructions. RNA transcription was performed *in vitro* with the HiScribe T7 High Yield RNA Synthesis Kit (New England BioLabs). DNA substrates containing the target cleavage site (479 base pairs, [Figure S13](#)) were produced by polymerase chain reaction (PCR) using mouse genomic DNA as a template and primer pairs 5'CAAAGAGCTCTCGTCCAGT3' and 5' ATGGACTCCAGGACCCAAGA3' followed by a column purification using the NucleoSpin® Gel and PCR Clean-up Kit (Macherey-Nagel). For the *in vitro* activity assay, RNP formation was achieved by incubating 3 μM *GeoCas9* (WT, K267E, R332A, or K267E/R332A mutant) and 3 μM gRNA at either 37 °C, 60 °C, 75 °C, or 85 °C for 30 minutes in a reaction buffer of 20 mM Tris, 100 mM KCl, 5 mM MgCl₂, 1 mM DTT, and 5% glycerol at pH 7.5. The 10 μL cleavage reactions were set up by mixing RNP at varying concentrations with 149 nanograms of PCR products on ice followed by incubation at 37 °C for 30 minutes. The reaction was quenched with 1 μL of proteinase K (20 mg/mL) and subsequent incubation at 56 °C for 10 minutes. 6x DNA loading buffer was added to each reaction and 10 μL of reaction mixture per lane was loaded onto an agarose gel. DNA band intensity measurements were carried out with ImageJ.

For *in vitro* off-target activity assays, RNP formation was achieved by incubating 10 μM *GeoCas9* (WT, K267E, R332A, or K267E/R332A mutant) and 10 μM gRNA at 37 °C for 30 minutes in the reaction buffer described above. The 10 μL cleavage reactions were set up by mixing 1 μM RNP with 150 nanograms of PCR products (off-target DNA sequences listed in [Table S2](#)) on ice followed by incubation at 37 °C for varying time points. The reaction was quenched with 1 μL of proteinase K (20 mg/mL) and subsequent incubation at 56 °C for 10 minutes. 6x DNA loading buffer was added to each reaction and 10 μL of reaction mixture per lane was loaded onto an agarose gel. DNA band intensity measurements were carried out with ImageJ. WT and HiFi *SpyCas9* control proteins were purchased from Integrated DNA Technologies (IDT, cat. No. 108158 and No. 108160, respectively), as was the associated *SpyCas9* gRNA, Alt-R™ CRISPR-Cas9 gRNA, with an RNA spacer sequence complementing 5'-TGGACAGAGCCTTCTTCTTC-3'. The on-target and off-target DNA sequences used for the *SpyCas9* *in vitro* cleavage assay can be found in [Table S3](#).

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Additional information

Author Contributions

HBB and **ALK** produced *GeoRec2*, *GeoRec1*, *GeoRec*, and *GeoCas9* proteins and RNAs, conducted the NMR and biophysical experiments, analyzed the data, and wrote the original draft of the manuscript. **AMD** solved the X-ray crystal structure of *GeoRec2*. **CP** carried out MD simulations and analyzed the data. **ZF** and **JL** conducted *GeoCas9* functional assays and analyzed the data. **GP** supervised the MD studies and obtained funding. **GJ** supervised collection of X-ray crystallographic data. **GPL** conceived the study, supervised collection of NMR spectroscopic data, obtained funding, and wrote the original draft. The final manuscript was written and edited with contributions from all authors.

Additional files

Supplementary Information 

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Reviewer #1 (Public review):

Summary:

In this study from Belato, Knight and co-workers, the authors investigated the Rec domain of a thermophilic Cas9 from *Geobacillus stearothermophilus* (GeoCas9). The authors investigated three constructs, two individual subdomains of Rec (Rec1 and Rec2) and the full Rec domain. This domain is involved in binding to the guide RNA of Cas9, as well as the RNA-DNA duplex that is formed upon target binding. The authors performed RNA binding and relaxation experiments using NMR for the wild-type domain as well as two-point mutants. They observed differences in RNA binding activities as well as the flexibility of the domain. The authors also performed molecular dynamics and functional experiments on full-length GeoCas9 to determine whether these biophysical differences affect the RNA binding or cleavage activity. Although the authors observed some changes in the thermal stability of the mutant GeoCas9-gRNA complex, they did not observe substantial differences in the guide RNA binding or cleavage activities of the mutant GeoCas9 variants.

Overall, this manuscript provides a detailed biophysical analysis of the GeoCas9 Rec domain. The NMR assignments for this construct should prove very useful, and can serve as the basis for future similar studies of GeoCas9 Rec domain mutants. While the two mutants tested in the study did not produce significant differences from wild-type GeoCas9, the study rules out the possibility that analogous mutations can be translated between type II-A and II-C Cas9 orthologs. Together, these findings may provide the grounds for future engineering of higher fidelity variants of GeoCas9

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Reviewer #2 (Public review):

The manuscript from Belato et al., used advanced NMR approaches and a mutagenesis campaign probe the conformational dynamics of the recognition lobe (Rec) of the CRISPR Cas9 enzyme from *G. stearothermophilus* (GeoCas9). Using truncated and full-length constructs they assess the impacts of two different point mutations have on the redistribution and timescale of these motions and assess gRNA recognition and specificity. Single point mutations in the Rec domain in a Cas9 from a related species had profound impacts on- and off-target DNA editing, therefore the authors reasoned analogous mutations in GeoCas9

would have similar effects. However, despite a redistribution of local motions and changes in global stability, their chosen mutations had little impact on DNA editing in the context of the full-length enzyme.

In their revised manuscript, the authors were highly responsive to the reviewer's comments incorporating new experimental results including molecular dynamics simulations and RNA binding data using full-length GeoCas9, as well as reframing their discussion and conclusions in consideration of the new data. They were receptive to suggestions for clarification in both the text and methods section. With these changes, the manuscript has been significantly improved.

Their studies highlight the species-specific complexity of interdomain communication and allosteric mechanisms used by these multi-domain endonucleases. The noted strengths of the article remain, and despite the negative results, their approach will garner interest from investigators interested in understanding how the activity and specificity of these enzymes can be engineered to tune activity and limit off-target cleavage by these enzymes. Generally, the manuscript highlights the challenges of studying the effect of allosteric networks on protein function, particularly in multidomain proteins, and thus will be of broad interest to the community.

<https://doi.org/10.7554/eLife.99275.3.sa2>

Reviewer #3 (Public review):

The authors explore the role of Rec domains in a thermophilic Cas9 enzyme. They report on the crystal structure of part of the recognition lobe, its dynamics from NMR spin relaxation and relaxation-dispersion data, its interaction mode with guide RNA, and the effect of two single-point mutations hypothesised to enhance specificity. They find that mutations have small effects on Rec domain structure and stability but lead to significant rearrangement of micro- to milli-second dynamics which does not translate into major changes in guide RNA affinity or DNA cleavage specificity, illustrating the inherent tolerance of GeoCas9. The work can be considered as a first step towards understanding motions in GeoCas9 recognition lobe, although no clear hotspots were discovered with potential for future rational design of enhanced Cas9 variants.

Strengths:

- Detailed biophysical and structural investigation, despite a few technical limitations inherent with working with complex targets, provides converging evidence that molecular dynamics embedded in the recognition lobes allow GeoCas9 to operate on a broad range of substrates.
- Since the authors and others have shown that substrate specificity is dictated by equivalent hotspot mutations in other Cas9 variants, we are one step closer to understanding this phenomenon.

Weaknesses:

- Since the mutations investigated here do not significantly affect substrate binding or enzymatic activity, it is difficult to rationalize anything for enzyme engineering at this point.
- Further investigation of the determinants of the observed dynamic modes, and follow-up with rationally designed mutations would hopefully allow to create a real model of the mechanism, but I do understand that this goes beyond the scope of this study.

<https://doi.org/10.7554/eLife.99275.3.sa1>

Author response:

The following is the authors' response to the previous reviews

Responses to final minor critiques following initial revision

Reviewer #1 (Recommendations for the authors):

The authors have generally done an excellent job of addressing my and the other reviewers' concerns. I have a few additional concerns that the authors could consider addressing through changes to the text:

We thank the Reviewer for this assessment and are glad to have addressed the major points.

- Regarding the gRNA used for NMR studies, I thank the authors for adding additional rationale for their design of the RNA used. However, I still believe that it is misleading to term this RNA as a "gRNA", given that it is mainly composed of a sequence that is arbitrary (the spacer) and the sections of the gRNA that are constant between all gRNAs are truncated in a way that removes secondary structure that is likely essential for specific contacts with the Rec domains. I do not believe the authors need to make alterations to any of their experiments. However, I do think their description of the "gRNA" should be updated to properly reflect that this RNA lacks any of the secondary structure present in a typical gRNA, much of which is necessary to confer specificity of binding between GeoCas9 and the gRNA. As mentioned in my previous review, this may be best achieved by adding a cartoon of the secondary structure of the full-length gRNA and highlighting the region that was used in the truncated "gRNA".

We understand the Reviewer's point. For any experiment in which the gRNA was truncated (i.e. NMR or some MST studies), we have clarified the text and no longer call it a "gRNA." We state initially that it is a portion of the gRNA and then call it simply an "RNA."

For experiments using the full-length constructs, we have kept the term "gRNA," as it remains appropriate.

We have also added a final Supplementary figure (S12) showing the structures of the truncated and full-length RNAs used, based on the _Geo_Cas9 cryo-EM structure and predicted with RNAfold.

- Lines 256-257: "The ~3-fold decrease in Kd..." I believe the authors are discussing the Kd's of the mutants relative to WT, in which case the Kd increased. Also, the fold-change appears closer to 2fold than to 3-fold.

Yes, the Reviewer makes a good catch. We have corrected this.

- Lines 407-408: "The mutations also diminished the stability of the full-length GeoCas9 RNP complex." This statement seems at odds with the authors' conclusions in the Results section that the full-length GeoCas9 variants had comparable affinities for the gRNAs (lines 376-382)

We agree that this seems contradictory. In the absence of full-length structures for all variants, we can't definitively state what causes this. It could be that the mutation has an interesting allosteric effect on structure that does not affect RNA binding but induces the Cas9 protein to simply fall apart at lower temperatures, rendering the binding interaction moot. We have added a statement to this section.

- The authors chose to keep "SpCas9" for consistency with their prior work and the work of many several others, including Doudna et al and Zhang et al. However, I will note that their publications on GeoCas9, the Doudna lab did use SpyCas9 to ensure consistent nomenclature within the publications.

We have made the change to "_Spy_Cas9"

Reviewer #3 (Recommendations for the authors):

The authors clearly answered most of my concerns. I still have some technical questions about the analysis of CPMG-RD data but the numbers provided now seem to make sense. While I still think that crystal structures of the point mutant would make the conclusions more "bullet proof", I do appreciate the work associated with this and consider that the manuscript can be published as is.

We agree that additional magnetic fields could allow for additional models of CPMG data fitting and that additional crystal structures of the mutants could add to the conclusions. We appreciate the Reviewer recognizing the balance of the current results and potential future studies in signing off on publication.

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